

Tarun Joseph

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Applied AI/ML Engineer

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TECHNICAL SKILLS

Programming: Python, R, Java, SQL, Git | **Generative AI & Agent Tools:** Transformers, FastAPI, Streamlit/Gradio, OpenAI APIs, Claude Code, GitHub Copilot | **Frameworks & Cloud:** LangChain, TensorFlow, PyTorch, GCP, AWS, Azure

PROFESSIONAL EXPERIENCE

- ❖ **Applied AI Project Management Intern | Vector Institute** May 2026 – Present
 - **Program Design & Process Improvement:** Support delivery of applied AI projects for Canadian SMEs by scoping technical requirements and translating research into structured, executable project plans. Automate daily workflows and processes to enable more agile and productive process workflows.
 - **Stakeholder Management:** Serve as liaison between Vector Institute researchers and industry sponsors, translating technical constraints into clear project roadmaps and deliverable timelines.
 - **Technical Translation:** Facilitate cross-functional workflows by communicating AI requirements and deployment considerations to non-technical partners, ensuring alignment from scoping through delivery.
- ❖ **Machine Learning Engineer | Tata Consultancy Services (TCS)** January 2023 – May 2025
 - **AI Prototyping:** Deployed GPT-3.5 and Llama 3-powered RAG chatbots and AI agents for TCS' Edtech platform, benefiting 600,000 associates globally.
 - **Workflow Automation:** Designed a PoC and implemented an MVP for support ticket automation using open-source LLMs, successfully eliminating redundant manual efforts and iterating model outputs based on stakeholder feedback across three sprint cycles thereby saving over five hours per developer per week
 - **Client Advisory & Teaching:** Led deep-skilling GenAI bootcamps for 10,000+ associates, translating model capabilities into practical workflows for non-technical teams, driving a 13% increase in platform engagement.
 - **Optimization:** Automated redundant manual tasks across developer teams using Generative AI tools, reducing overall manual effort by 30%.

PROJECTS

- ❖ **AI Digital Twin Agent | LangChain, OpenAI APIs, Python** June 2026
 - Built a multi-tool AI agent that combines vector-store memory, semantic routing, and live API calls to interpret unstructured requests and autonomously execute multi-step workflows.
 - Evaluated and debugged end-to-end agent reasoning, retrieval, and tool-use traces; refined orchestration logic to improve reliability and deliver a deployable proof of concept for real-world automation.
- ❖ **Support Ticket Automation Agent | Llama 3, Python, Open-Source LLMs** March 2025
 - Designed a proof-of-concept (PoC) and built an MVP for an autonomous IT support agent using **Llama 3** with tool-calling capabilities, eliminating repetitive ticket resolution without human intervention.
 - Delivered measurable impact at enterprise scale across a platform used by **600,000 employees**, saving over **five hours** per developer per week.
- ❖ **Real-Time ASL Translator | PyTorch, Computer Vision, Deep Learning** April 2026
 - Led a four-person team to design and build a real-time **American Sign Language recognition** system using computer vision and deep learning, achieving **65%** classification reliability.

TEACHING EXPERIENCE

- ❖ **Graduate Teaching Assistant | University of Toronto** September 2025 – April 2026
 - **MIE429 AI/ML Capstone Design:** Mentored 45 final-year students in designing AI prototypes, stepping in to resolve technical roadblocks during meetings with industry clients to ensure successful project delivery.
 - **MIE1628 (Cloud-based Big Data Analytics):** Supervised labs for 500+ students, teaching Python-based data workflows and debugging code under high-volume, fast-paced conditions.

EDUCATION & CERTIFICATIONS

- ❖ **MEng, Industrial Engineering | University of Toronto, Canada** September 2025 – Present
 - **Relevant Courses:** Deep Learning, Data Science & Statistics, Natural Language Processing, HR Management
 - **Emphasis:** Data Analytics & Machine Learning, ELITE | GPA: 3.8/4
- ❖ **B. Tech, Industrial Engineering | College of Engineering Trivandrum, India** August 2017 – June 2021
- ❖ **Certifications:** Google: Generative AI Leader Certification (2026) | AWS Certified AI Practitioner (2024) | Microsoft Certified Azure Data Scientist Associate (2024) | AWS Certified Cloud Practitioner (2023)